

Solutions to selected problems in field theory

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1 Introduction

These notes contain selected solutions and comments on problems in field theory from Jacobson's *Basic Algebra*. They grew out of my work on the weekly assignments for the Algebra II course at The Institute of Mathematical Sciences, Chennai. The notes are informal and may contain errors. They are not meant to be a complete or official solution manual. If you find any mistakes or ambiguities, please feel free to write to me at haritr@imsc.res.in.

2 Splitting Field of a Polynomial

Problem 1: Let $\omega = \cos\pi/6 + i \sin\pi/6$. Show that $[\mathbb{Q}(\omega) : \mathbb{Q}] = 4$ and determine the minimal polynomial of ω over $\mathbb{Q}$.

Solution: Clearly $\omega^6 = -1 \neq 0$ and $\omega^6 + 1 = 0$. Now

$$x^6 + 1 = (x^2 + 1)(x^4 - x^2 + 1)$$

and ω is a root of $(x^4 - x^2 + 1)$. We claim that $f(x) = (x^4 - x^2 + 1)$ is irreducible over $\mathbb{Q}$. Since $f(x)$ has no roots in $\mathbb{Q}$, if it was irreducible then it must be the product of 2 monic quadratic polynomials say

$$(x^4 - x^2 + 1) = (x^2 + ax + b)(x^2 + cx + d) \quad a, b, c, d \in \mathbb{Q}$$

Expanding and comparing coefficients, we get

$$a + c = 0, \quad \text{i.e. } c = -a$$

$$ac + b + d = -1$$

$$ad + bc = 0$$

$$bd = 1$$

Substituting $c = -a$ and $b = 1/d$ in the above equations and reducing, we get $a = 0$ or $d^2 = 1$. If $a = 0$, then

$$f(x) = x^4 + (b + d)x^2 + bd \tag{1}$$

implying $b + d = -1$ and $bd = 1$, i.e., b, d are $(-1 \pm \sqrt{-3})/2 \notin \mathbb{Q}$. This is a contradiction. Similarly if $d^2 = 2$, then we have $b = 1$, $ac + b + d = -1$, that is $-a^2 + 2 = -1$, implying $a \notin \mathbb{Q}$. This is also not possible. Therefore $f(x)$ is irreducible over $\mathbb{Q}$. i.e.,

$$[\mathbb{Q}(\omega) : \mathbb{Q}] = \deg(\min_{\mathbb{Q}}(\omega)) = 4.$$

Problem 2: Let $E = F(u)$ where u is algebraic of odd degree. Show that $E = F(u^2)$.

Solution: Note that $F(u^2) \subseteq F(u)$. Since u is algebraic over F , $[F(u) : F] < \infty$. Suppose $F(u^2) \neq F(u)$, then $[F(u) : F(u^2)] = 2$ since $x^2 - u^2$ would be the minimal polynomial of u over $F(u^2)$. But we know that $[F(u) : F] = [F(u^2) : F(u)][F(u^2) : F] = 2 \times [F(u^2) : F]$. This implies 2 divides $[F(u) : F]$, which is a contradiction since it is odd by the hypothesis. Therefore

$$F(u^2) = F(u).$$

Problem 3: Let E be an extension field of F which is algebraic over F . Show that every subring E/F is a subfield. Hence prove that any subring of a finite dimensional extension field E/F is a subfield.

Solution: Let R be a subring of E/F . Let $a \in R$ with $a \neq 0$. Then $a^{-1} \in E$ and

satisfies a polynomial equation of the form

$$a^{-m} + a_{m-1}a^{-(m-1)} + \cdots + a_0 = 0$$

with $a_i \in F$ for $i = 0, 1, \dots, m-1$. Multiply the above equation by $a^{m-1} \in R$ to get

$$a^{-1} = -(a_{m-1} + \cdots + a_0 a^{m-1}).$$

Since $a, a_i \in R, a^{-1} \in R$. Therefore R is a field. Now to prove that any subring of a finite dimensional extension E/F is a subfield it is enough to show that E is algebraic over F . Assume the dimension of E over F is n . Let $x \in E$. Then since the dimension of E over F is n , $\{1, x, x^2, \dots, x^n\}$ is a linearly independent set. Therefore there exists a polynomial

$$p(x) = a_0 + a_1x + \cdots + a_nx^n$$

with not all $a_i = 0$ implying x is algebraic over F and thus E is algebraic over F .

Problem 4: Let $E = F(u)$, u transcendental, and let $K \neq F$ be a subfield of E/F . Show that u is algebraic over K .

Solution: Since $K \neq F$, there exists $v \in K \setminus F$ such that $v = \frac{f(u)}{g(u)}$ where $f(x), g(x) \in F[x]$ with $g(x) \neq 0$, $(g(x), f(x)) = 1$. Let

$$p(x) = vg(x) - f(x). \tag{2}$$

Then $p(x) \in K[x]$ and $p(u) = 0$. If in case $p(x) = 0$ with $g(0) \neq 0$, then $f(0) = vg(0) \Rightarrow v = f(0)g(0)^{-1} \in F$ which is not possible as $v \notin F$. If in case $p(x) = 0$ with $g(0) = 0$, then $p(0) = f(0) = 0$. i.e., x is a common factor of f and g . But this is a contradiction as we assumed that $(g(x), p(x)) = 1$. Therefore we have $p(x) \neq 0$. Then from (2), $p(x) \in K[x]$ and $p(u) = 0$. i.e., u is algebraic over K .

Problem 5: Let E be an extension field of the field F such that $[E : F] < \infty$, for any two subfields E_1 and E_2 containing F , either $E_1 \subset E_2$ or $E_2 \subset E_1$. Show that E has a primitive element over F .

Solution: We know that if $[E : F] < \infty$, then E is algebraic over F . Let d be the highest degree of the polynomial in the collection of the minimal polynomials of elements of E . Let $\alpha \in E$ be such that degree of α over F is d . We claim that $E = F(\alpha)$. If not, there exists $\beta \in E \setminus F(\alpha)$. Then by the conditions given in the question, $F(\beta) \subset F(\alpha)$ or $F(\alpha) \subset F(\beta)$. Since $\beta \notin F(\alpha)$, it follows that $F(\alpha) \subset F(\beta)$. Then the degree of β over F is greater than the degree of α over F which is d . But this contradicts the maximality of d . Therefore $E = F(\alpha)$.

Problem 6: Construct a splitting field over $\mathbb{Q}$ of $x^5 - 2$. Find its dimensionality over $\mathbb{Q}$.

Solution: Let $p(x) = x^5 - 2$. The roots of $p(x)$ are $2^{1/5}, 2^{1/5}\zeta, 2^{1/5}\zeta^2, 2^{1/5}\zeta^3, 2^{1/5}\zeta^4$ where ζ is a primitive fifth root of unity. Then the splitting field of $p(x)$ over $\mathbb{Q}$ is $\mathbb{Q}(2^{1/5}, 2^{1/5}\zeta, 2^{1/5}\zeta^2, 2^{1/5}\zeta^3, 2^{1/5}\zeta^4)$. Call the above field L . It is obvious that $L = \mathbb{Q}(2^{1/5}, \zeta)$. Now to find $[L : \mathbb{Q}]$. We know that by tower rule,

$$[L : \mathbb{Q}] = [\mathbb{Q}(2^{1/5}, \zeta) : \mathbb{Q}(2^{1/5})][\mathbb{Q}(2^{1/5}) : \mathbb{Q}] = [\mathbb{Q}(2^{1/5}, \zeta) : \mathbb{Q}(\zeta)][\mathbb{Q}(\zeta) : \mathbb{Q}] \quad (3)$$

But $[\mathbb{Q}(\zeta) : \mathbb{Q}] = 4$ since $x^4 + x^3 + x^2 + x + 1$ is the minimal polynomial of ζ over $\mathbb{Q}$. Also, $[\mathbb{Q}(2^{1/5}) : \mathbb{Q}] = 5$ since $x^5 - 2$ is the minimal polynomial of $2^{1/5}$ over $\mathbb{Q}$ ($2^{1/5}$ is a root of the polynomial and the polynomial is irreducible over $\mathbb{Q}$ by the Eisenstein's criteria for prime 2). Therefore, we have that 4 and 5 divide $[L : \mathbb{Q}]$ from (3). Since 4 and 5 are relatively prime, $4 \times 5 = 20$ divides $[L : \mathbb{Q}]$. i.e., $[L : \mathbb{Q}] \leq 20$. On the other hand, note that $[\mathbb{Q}(2^{1/5}, \zeta) : \mathbb{Q}(2^{1/5})] \leq 4$ since ζ satisfies the polynomial $x^4 + x^3 + x^2 + x + 1 \in \mathbb{Q}(2^{1/5})[x]$. Therefore, $[L : \mathbb{Q}] \leq 4 \times 5 = 20$ (from (3)). Thus, $20 \leq [L : \mathbb{Q}] \leq 20 \Rightarrow [L : \mathbb{Q}] = 20$.

Problem 7: Determine a splitting field over $\frac{\mathbb{Z}}{(p)}$ of $x^{p^e} - 1$, $e \in \mathbb{N}$.

Solution: Since $\frac{\mathbb{Z}}{(p)}$ has characteristic p ,

$$(x - 1)^{p^e} = x^{p^e} - 1.$$

Thus the polynomial splits over $\frac{\mathbb{Z}}{(p)}$ itself, and its splitting field is $\frac{\mathbb{Z}}{(p)}$.

Problem 8: Let E be an extension field of F such that $[E : F] = n < \infty$. Let K be any extension field of F . Show that the number of monomorphisms of E/F into K/F does not exceed n .

Solution: Let $\eta : F \rightarrow F'$ be an isomorphism of the fields F and F' . Let E and K be two extension fields of F and F' respectively. We want to show that the number of monomorphisms from E to K extending η does not exceed n . We induct on n . When $[E : F] = 1$, then $E = F$ and there is only one monomorphism from E to K namely η itself. Now assume that the statement is true for all fields E and F with $[E : F] < n$. Now let $[E : F] = n > 1$. Then there exists $r \in E \setminus F$ and $[F(r) : F] > 1 \Rightarrow [E : F(r)] < n$. Let ζ be a monomorphism from E/F to K/F' (i.e., $\zeta|_F : F \rightarrow F'$ is an isomorphism. Then $\phi = \zeta|_{F(r)}$ is a monomorphism from $F(r)/F$ to $\phi(F(r))/F'$. Therefore every monomorphism of E/F to K/F' is an extension of an isomorphism from $F(r)$ to a subfield of K .

Let $\psi : F(r) \rightarrow K' \subseteq K$ be an isomorphism. By induction hypothesis, the number of monomorphisms from E to K extending ψ is less than or equal to $[E : F(r)]$. By the lemma after theorem 4.4 in Jacobson, the number of monomorphisms from $F(r)$ to a subfield of K extending the isomorphism of F and F' is $\leq [F(r) : F]$. Thus the number of monomorphisms from E to K extending $\eta : F \rightarrow F'$ is less than or equal to $[E : F(r)][F(r) : F] = [E : F]$. Now taking $F = F'$ and $\eta = Id$, the result follows.

3 The Galois Group, The Fundamental Theorem of Pairing

Problem 9: Show that in the subgroup-intermediate subfield correspondence given in the fundamental theorem of Galois theory, the subfield corresponding to the intersection of two subgroups H_1 and H_2 is the subfield generated by the corresponding intermediate fields, and the intersection of two intermediate fields K_1 and K_2 corresponds to the subgroup generated by the corresponding subgroups.

Solution: Let $K_1 = \text{Inv}H_1$ and $K_2 = \text{Inv}H_2$. We need to prove that $\text{Inv}(H_1 \cap H_2) = K_1K_2$. Initially we claim that $H_1 \cap H_2 = \text{Gal}(E/K_1K_2)$. Let $\sigma \in H_1 \cap H_2$. Then σ fixes K_1 and K_2 so that σ fixes K_1K_2 , i.e., $\sigma \in \text{Gal}(E/K_1K_2)$. Also, if $\sigma \in \text{Gal}(E/K_1K_2)$, then σ fixes $K_1K_2 \Rightarrow \sigma$ fixes K_1 and $K_2 \Rightarrow \sigma \in H_1 \cap H_2$. Therefore we get $H_1 \cap H_2 = \text{Gal}(E/K_1K_2)$. Thus $\text{Inv}(H_1 \cap H_2) = \text{Inv}(\text{Gal}(E/K_1K_2)) = K_1K_2$ (by theorem 4.7 of Jacobson). E is the splitting field of a separable polynomial $f(x) \in K_1K_2[x]$.

For the second part, we need to show that $\text{Gal}(E/K_1 \cap K_2) = \text{Gal}(E/K_1)\text{Gal}(E/K_2)$. Set $H_1 = \text{Gal}(E/K_1)$ and $H_2 = \text{Gal}(E/K_2)$. We claim that $K_1 \cap K_2 = \text{Inv}(H_1H_2)$. If $x \in K_1 \cap K_2$, and $\sigma \in H_1H_2$, then we have $\sigma(x) = x$ as σ is an F -linear combination of $\sigma_i\tau_i$ where $\sigma_i \in H_1$ and $\tau_i \in H_2$. Therefore $x \in \text{Inv}(H_1H_2)$. Now, we have $\text{Inv}(H_1H_2) \subseteq K_1$ and $\text{Inv}(H_1H_2) \subseteq K_2$ implying $\text{Inv}(H_1H_2) \subseteq K_1 \cap K_2$. Therefore $K_1 \cap K_2 = \text{Inv}(H_1H_2)$, i.e., $\text{Gal}(E/K_1 \cap K_2) = \text{Gal}(\text{Inv}(H_1H_2)) = H_1H_2$.

Problem 10: Suppose E, F, G are as in fundamental theorem (Jacobson), and E is generated by two intermediate extensions K and L such that $K \cap L = F$ and L/F is normal. Let $N = \text{Gal}(E/L)$, $H = \text{Gal}(E/K)$. Show that N is normal in G , $H \cap N = 1$ and $G = HN$, so G is the semi direct product of N and H . Show also that if K/F and L/F are normal then $G = H \times N$.

Solution: Let $N = \text{Gal}(E/L)$ and $H = \text{Gal}(E/K)$. By fundamental theorem of Galois theory, we know that N is normal in G iff $\text{Inv}N$ is normal over F . But here, $\text{Inv}N = \text{Inv}(\text{Gal}(E/L)) = L$ which is normal over F . Therefore N is normal in G .

The field $\text{Inv}(H \cap N)$ is the subfield generated by $\text{Inv}H$ and $\text{Inv}N$ by the previous problem. Note that $\text{Inv}H = K$ and $\text{Inv}N = L \Rightarrow \text{Inv}(H \cap N) = KL = E \Rightarrow \text{Gal}(\text{Inv}(E/(H \cap N))) = \text{Gal}(E/E) \Rightarrow H \cap N = 1$.

Now by the previous problem, we have $\text{Gal}(E/K \cap L)$ is the subgroup generated by $\text{Gal}(E/K)$ and $\text{Gal}(E/L)$. Let $\text{Gal}(E/K) = H$ and $\text{Gal}(E/L) = N$. The fact $N \triangleleft G \Rightarrow HN$ is a subgroup of G . Also HN is the smallest subgroup of G containing H and N . Therefore $\text{Gal}(E/K \cap L) = HN$, i.e., $\text{Gal}(E/F) = G = HN$.

Now recall the group theory proposition; if H and N are two normal subgroups of

a group G with $H \cap N = 1$ and $HN = G$, then $G \cong H \times N$. Here, if K/F and L/F are normal, then by the fundamental theorem of Galois theory, H and N are normal subgroups of G . This together with the above proposition and the above parts imply that $G \cong H \times N$.

Problem 11: Let $E = \mathbb{Q}(r)$ where $r^3 + r^2 - 2r - 1 = 0$. Verify that $r' = r^2 - 2$ is also a root of $x^3 + x^2 - 2x - 1 = 0$. Determine $\text{Gal}(E/\mathbb{Q})$. Show that E is normal over $\mathbb{Q}$.

Solution: Let $f(x) = x^3 + x^2 - 2x - 1$ with r as a root. Putting $x = r^2 - 2$ in the equation yields zero. We know that the sum of roots of polynomial of degree n is the negative of the coefficient of x^{n-1} . Therefore if λ is the third root of $f(x)$, then we have $r + r^2 - 2 + \lambda = -1 \Rightarrow \lambda = 1 - r - r^2$. Note that the three roots are distinct since for example if $r = r^2 - 2$, then $r = 2, -1$ which are not roots of $f(x)$. Other cases also can be ruled out in the same way. Now if p/q is a root of $f(x)$, then by rational root theorem, $p| -1$ and $q|1 \Rightarrow p/q = \pm 1$. But it is clear that ± 1 are not roots of $f(x)$. Therefore $f(x)$ is a polynomial of degree three with no roots in $\mathbb{Q}$. i.e., $f(x)$ is irreducible over $\mathbb{Q}$. Since all the roots of $f(x)$ lies in $\mathbb{Q}(r)$, $\mathbb{Q}(r)$ is a splitting field of $f(x)$ over $\mathbb{Q}$. i.e., $\mathbb{Q}(r)$ is the splitting field of a separable polynomial. Therefore $E = \mathbb{Q}(r)$ over $\mathbb{Q}$ is normal. Therefore

$$|\text{Gal}(\mathbb{Q}(r)/\mathbb{Q})| = [\mathbb{Q}(r) : \mathbb{Q}] = 3.$$

Since there is only group of order 3, we get $\text{Gal}(E/\mathbb{Q}) \cong \mathbb{Z}/3\mathbb{Z}$.

Problem 12: Let F be a field of characteristic p , a an element not of the form $b^p - b$, $b \in F$. Determine the Galois group over F of a splitting field of $x^p - x - a$.

Solution: First we claim that if α is a root of $f(x) = x^p - x - a$, then $\alpha + i$, $i = 1, 2, \dots, p-1$ are also roots of $f(x)$. Then all the roots of $f(x)$ will be distinct. Note that

$$(\alpha + i)^p - (\alpha + i) - a = \alpha^p + i^p - \alpha - i - a = i^p - i = 0.$$

We also claim that $f(x)$ is irreducible over F . If $f(x)$ was reducible, then assume $f(x) = g(x)h(x)$ with $h(x), g(x) \in F[x]$ and $1 < \deg h(x) < p$. Since $F[x]$ is a UFD and we know all the roots of $f(x)$, we can assume that $h(x) = (x - (\alpha + 1)) \cdots (x - (\alpha + n))$. Then the coefficient of x^{n-1} in $h(x)$ is $-n\alpha + (1 + 2 + \cdots + n)$ and it belongs to F . This implies that $n\alpha \in F$ which in turn implies $\alpha \in F$. But in the question, it is assumed that $f(x)$ has no roots in F . This is a contradiction. Therefore $f(x)$ is irreducible over F . Also it must be clear that the splitting field of $f(x)$ over F is $F(\alpha)$. Now, $F(\alpha)$ is the splitting field of a separable polynomial $f(x)$, $F(\alpha)/F$ is normal and $|\text{Gal}(F(\alpha)/F)| = [F(\alpha) : F] = p$. Therefore $\text{Gal}(F(\alpha)/F) \cong \mathbb{Z}/p\mathbb{Z}$.

Problem 13: Show that $E = \mathbb{Q}(\sqrt{3}, \sqrt{2}, u)$, where $u^2 = (9 - 5\sqrt{3})(2 - \sqrt{2})$ is normal. Determine $\text{Gal}(E/\mathbb{Q})$.

Solution: By an equivalent condition for normality, to prove that E is normal over $\mathbb{Q}$, it is enough to show that every embedding of E in $\bar{\mathbb{Q}}$ (algebraic closure) is an automorphism of E . Since an automorphism of field maps algebraic elements to its conjugates, an automorphism of E can map $\sqrt{2}$ to $\pm\sqrt{2}$ and $\sqrt{3}$ to $\pm\sqrt{3}$. From the definition of u^2 , it is clear that the image of u^2 under any automorphism of E is determined by the images of $\sqrt{2}$ and $\sqrt{3}$. Now let us check the possible automorphisms of E with their images of $\sqrt{2}$ and $\sqrt{3}$ to get an idea about the possible images of u . The possible automorphisms are

$$\begin{array}{c|c|c|c} \sigma_1 & \sigma_2 & \sigma_3 & \sigma_4 \\ \hline \sqrt{2} \rightarrow \sqrt{2} & \sqrt{2} \rightarrow \sqrt{2} & \sqrt{2} \rightarrow -\sqrt{2} & \sqrt{2} \rightarrow -\sqrt{2} \\ \sqrt{3} \rightarrow \sqrt{3} & \sqrt{3} \rightarrow -\sqrt{3} & \sqrt{3} \rightarrow \sqrt{3} & \sqrt{3} \rightarrow -\sqrt{3} \end{array}$$

Let us check the image of u^2 under each of the maps above:

$$\begin{array}{c|c} \sigma_1(u^2) = (9 - 5\sqrt{3})(2 - \sqrt{2}) & \sigma_2(u^2) = (9 + 5\sqrt{3})(2 - \sqrt{2}) \\ \sigma_3(u^2) = (9 - 5\sqrt{3})(2 + \sqrt{2}) & \sigma_4(u^2) = (9 + 5\sqrt{3})(2 + \sqrt{2}) \end{array}$$

If σ'_i s are automorphisms, then since $\sigma_i(u^2) = \sigma_i(u)^2$, the possible images of u are the square roots of the images of u^2 . There are 8 possible choices for the image of u namely $\pm u, \pm \frac{\sqrt{6}u}{9-5\sqrt{3}}, \pm(\sqrt{2}+1)u, \pm \frac{2\sqrt{3}}{u}$ (this we can easily compute by a simple calculation). Thus this implies there are potentially 8 possible automorphisms of E .

If σ is an embedding of E in $\bar{\mathbb{Q}}$, σ is completely determined by $\sigma(\sqrt{2}), \sigma(\sqrt{3})$, and $\sigma(u)$. Since each embedding maps $\sqrt{2}, \sqrt{3}$ and u to their conjugates only, by the above calculations, it must be clear that σ must be an automorphism of E . Therefore E is normal over $\mathbb{Q}$.

Now we want to show that $[E : \mathbb{Q}] = 8$. Since $x^2 - u^2 \in \mathbb{Q}(\sqrt{2}, \sqrt{3})[x]$, and u is a root of that polynomial, $[E : \mathbb{Q}(\sqrt{2}, \sqrt{3})]$ is either 1 or 2. If it was 1, then $u \in \mathbb{Q}(\sqrt{2}, \sqrt{3})$. In that case consider the automorphism say σ_3 of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ sending $\sqrt{2} \rightarrow -\sqrt{2}$ and $\sqrt{3} \rightarrow \sqrt{3}$. Then $\sigma_3^2 = 1$. But by the above calculations, $\sigma_3(u) = \pm(\sqrt{2}+1)u$. This would imply that $\sigma_3^2(u) = -u$ which is a contradiction. Therefore $[E : \mathbb{Q}(\sqrt{2}, \sqrt{3})] = 2$ and consequently $[E : \mathbb{Q}] = 8$. Since the extension over $\mathbb{Q}$ is normal, it is Galois. Therefore $|\text{Gal}(E/\mathbb{Q})| = [E : \mathbb{Q}] = 8$. This implies that all the 8 possible maps we described above are automorphisms of E . If we compute the squares of the automorphisms in $\text{Gal}(E/\mathbb{Q})$, the automorphism which sends u to $-u$ is the only element of order 2. Since the only groups of order 8 with single order two element are Q_8 and $\mathbb{Z}/8\mathbb{Z}$, Galois group is either of the two. But the Galois group cannot be $\mathbb{Z}/8\mathbb{Z}$ since E has 3 distinct subfields ($\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(u)$) which are quadratic extensions corresponding to three distinct subgroups of order 4 of the Galois group. Thus $\text{Gal}(E/\mathbb{Q}) \cong Q_8$.

4 Solvable and Nilpotent Groups

Problem 14: Define G^i by $G^1 = G$, $G^i = [G^{i-1}, G]$. The sequence of normal subgroups $G^1 \supset G^2 \supset \cdots$ is called the *lower central series* for G . G is called *nilpotent* if there exists an integer k such that $G^k = 1$. Show that if G is nilpotent, then it is solvable. Give an example to show that the converse does not hold.

Solution: We have $G^{(k)} = [G^{(k-1)}, G^{(k-1)}] \forall k \geq 1$ and $G^{(0)} = G$. Then by a simple induction, it is clear that $G^{(k)} \subset G^k$ for all k . If G is nilpotent, there exists an integer n such that $G^n = 1$. Then $G^{(n)} = 1$ and G is solvable.

But the converse need not hold. For example S_3 is solvable since $[S_3, S_3] = A_3$ and $[A_3, A_3] = 1$ (this follows from the fact $[G, G]$ is the smallest normal subgroup of G such that the quotient group is abelian.). But S_3 is not nilpotent since $S_3^2 = [A_3, S_3] = A_3$ and therefore $S_3^n = A_3$ for any $n \geq 2$ and never 1.

Explanation for $[A_3, S_3] = A_3$: Let $[x, \sigma] = x\sigma x^{-1}\sigma^{-1}$ with $x \in A_3, \sigma \in S_3$. Then obviously $[x, \sigma]$ is an even permutation and belong to A_3 . Now if $\sigma \in A_3$, then $\sigma = 1, (123)$ or (132) . Let $\sigma = (123)$. we have (123) and (132) are conjugates in S_3 since they have same cycle type. Therefore there exists $x \in S_3$ such that $(123)(123) = (132) = x(123)x^{-1}$. This implies that $(123) = (123)^{-1}x(123)x^{-1} \in [A_3, S_3]$. Similarly $(132) \in [A_3, S_3]$. Thus $[A_3, S_3] = A_3$.

Problem 15: Show that every element h of a nilpotent group G has the *Engel property*: there exists integer k such that $[[\cdots [g, h], h] \cdots h] = 1$ for every $g \in G$.

Solution: This follows easily from the fact if G is nilpotent, then there exists an integer k such that $G^k = 1$ and

$$\underbrace{[[\cdots [g, h], h] \cdots h]}_{k \text{ times}} \in G^k.$$

Problem 16: Prove that if H is a proper subgroup of a nilpotent group G , then the normalizer $N(H) \supsetneq H$.

Solution: Since G is nilpotent, there exists $n \in \mathbb{N}$ such that $G^n = 1$. Therefore we can find a $k \in \mathbb{N}$ such that $G^{k+1} \subseteq H$ but $G^k \not\subseteq H$. Choose an $x \in G^k \setminus H$. We claim that x is in $N(H)$. Let $h \in H$. Then $xhx^{-1}h^{-1} = [x, h] \in G^{k+1} \subseteq H \Rightarrow xhx^{-1} \in H \Rightarrow xHx^{-1} \subseteq H$. Now $h \in H \Rightarrow [x^{-1}, h] \in G^{k+1} \subseteq H \Rightarrow x^{-1}hx \in H \Rightarrow h = x\bar{h}x^{-1}$ for some $\bar{h} \in H$. Thus we get $H = xHx^{-1}$. Also it is obvious that $H \subseteq N(H)$. Hence H is a proper subgroup of $N(H)$.

Problem 17: Show that if G is a finite nilpotent group, then every Sylow subgroup is normal in G .

Solution: Let H be a Sylow p -subgroup of G . If $H = G$, we are done. If $H \neq G$, we have $N(H) \supsetneq H$. We claim that $N(N(H)) = N(H)$. Obviously $N(H) \subseteq N(N(H))$. If $g \in N(N(H))$, then $gN(H)g^{-1} = N(H)$. Also note that $gHg^{-1} \subseteq gN(H)g^{-1} = N(H)$. Therefore gHg^{-1} and H are two Sylow p -subgroups of $N(H)$. But H is normal in $N(H)$ since $N(H)$ is the maximal subgroup of G such that H is normal in that. Therefore by Sylow second theorem (Sylow subgroups are conjugate), we have $gHg^{-1} = H$, i.e., $g \in N(H)$. Thus we have $N(N(H)) = N(H)$. Then by the above exercise, $N(H) = G$. i.e., H is normal in G .

Problem 18: If G is a group, define the *upper central series* $1 \subset C_1 \subset C_2 \cdots$ by $C_1 = C(G)$, and C_i , the normal subgroup such that C_i/C_{i-1} is the center G/C_{i-1} . Show that a finite group G is nilpotent iff the upper central series ends in a finite number of steps with G . Prove that any p -group is nilpotent.

Solution: Let us fix some notations. Let $G^0 = G$, $G^1 = [G, G]$, $G^2 = [G^1, G] \cdots$ and $C_0 = 1$, $C_1, C_2, \cdots$ as in the question. Assume upper central series terminates with G . Let $k \geq 1$ such that $1 \subset C_1 \subset \cdots \subset C_k = G$. We claim that for $G^j \subseteq C_{k-j}$ for $j = 0, 1, \cdots, k$. When $j=0$, $G^0 = G \subseteq C_k = G$. Suppose $G^j \subseteq C_{k-j}$. Let $[g, h] \in G^{j+1}$ with $g \in G^j \subseteq C_{k-j}$ and $h \in G$. Since $g \in C_{k-j}$, $gC_{k-(j+1)} \in C_{k-j}/C_{k-(j+1)} = Z(G/C_{k-(j+1)})$. Therefore

$$\begin{aligned} gC_{k-(j+1)}hC_{k-(j+1)} &= hC_{k-(j+1)}gC_{k-(j+1)} \\ \Rightarrow ghC_{k-(j+1)} &= hgC_{k-(j+1)} \Rightarrow [g, h] \in C_{k-(j+1)} \end{aligned}$$

That is, $G^{j+1} \subseteq C_{k-(j+1)}$. Now if we put $j = k - 1$, then $G^k \subseteq C_0 = 1 \Rightarrow G^k = 1$. Hence G is nilpotent.

Conversely suppose that G is nilpotent with lower central series terminates with $G^k = 1$. Then by a similar argument above, we have $G^j \subseteq C_{k-j}$ for $j = 0, 1, \cdots, k$. Now put $j = 0$ to obtain $G^0 = G \subseteq C_k \Rightarrow C_k = G$. i.e., the upper central series stabilizes.

Let G be p -group. We want to show that G is nilpotent. Since the center of p -group is non-trivial, the center C_1 is also a p -group. Now if $C_1 = G$, then we are done. Otherwise G/C_1 is a p -group with non-trivial center C_2/C_1 . If $C_2 = G$, we are done, otherwise we get a subgroup C_3 of G such that $C_3 \supsetneq C_2$. Therefore we get a sequence $1 \subsetneq C_1 \subsetneq C_2 \cdots$. Since G is a finite group, this should terminate at some $C_k = G$. Then G is nilpotent.

Problem 19: Prove that a finite group is nilpotent iff it is a direct product of p -groups.

Solution: Let G be a finite group which is nilpotent. We claim that G is the direct product of its Sylow subgroups. Let $H_1, \cdots, H_n$ be the distinct Sylow subgroups of G . Then each G_i is a p -group. By the above exercise, H_i is normal in G for all i . Also

$H_i \cap H_j = 1$ for $i \neq j$. Now we have $|H_1 \cdots H_n| = |H_1| \cdots |H_n| = |G|$. Therefore we have $H_i \triangleleft G$, $H_i \cap H_j = 1$ for $i \neq j$ and $H_1 \cdots H_n = G$. Then $G \cong H_1 \times H_2 \times \cdots \times H_n$.

Conversely assume that G is a direct product of p -groups. By one of the above exercise, we know that each p -group is nilpotent. Now it is enough to show that the direct product of two nilpotent groups is nilpotent. Let H_1 and H_2 be nilpotent groups. Note that for all $i \geq n$, $(H_1 \times H_2)^i = H_1^i \times H_2^i$ (this we can prove easily by induction). Hence it must be clear by now that the $H_1 \times H_2$ is nilpotent.

5 Galois criterion for Solvability by Radicals

Problem 20: Let p be a prime unequal to the characteristic of the field F . Show that if $a \in F$, then $x^p - a$ is either irreducible in $F[x]$ or it has a root in F .

Solution: Let $f(x) = x^p - a \in F[x]$. If $f(x)$ is irreducible, then we are done. Assume $f(x)$ is reducible with a root α in some splitting field of $f(x)$. If $\zeta_1, \dots, \zeta_n$ are the p th roots of unity over F , then $\alpha\zeta_1, \dots, \alpha\zeta_p$ are the roots of $f(x)$. Therefore, $x^p - a = (x - \alpha\zeta_1) \cdots (x - \alpha\zeta_n)$ in $E[x]$ where E is some extension field of F . Since $f(x)$ is reducible, $f(x) = h(x)g(x)$ with $f(x), g(x) \in F[x]$ and $1 \leq \deg g(x) < n$. Then since $E[x]$ is a UFD, $g(x) = (x - \alpha\zeta_1) \cdots (x - \alpha\zeta_r)$ (after reordering, if necessary). This implies that $\zeta_1 \cdots \zeta_r \alpha^r \in F$. Set $\zeta = \zeta_1 \cdots \zeta_r$. Then $\zeta^p \in F$ and $\zeta \alpha^r \in F$. Since r and p are relatively prime, there exists integers s, t such that $rs + pt = 1$. Then $\zeta^s \alpha = \zeta^s \alpha^{rs+pt} = (\zeta \alpha^r)^s (\alpha^p)^t \in F$. Now $(\zeta^s \alpha)^p = (\zeta^p)^s \alpha^p = \alpha$. Therefore, $\zeta^s \alpha$ is a root of $f(x)$ and $\zeta^s \alpha \in F$. Hence $f(x)$ has a root in F .

Problem 21: Assume that $x^p - a, a \in \mathbb{Q}$, is irreducible in $\mathbb{Q}[x]$. Show that the Galois group of $x^p - a$ over $\mathbb{Q}$ is isomorphic to the group of transformations of $\mathbb{Z}/(p)$ of the form $y \rightarrow ky + l$ where, $k, l \in \mathbb{Z}/(p)$ and $k \neq 0$.

Solution: Let $f(x) = x^p - a$, be irreducible. Let ζ be a primitive p th root of unity, and $\alpha \in E$ be such that $\alpha^p = a$ where E is an extension field of $\mathbb{Q}$. Then all the roots of $f(x)$ are $\alpha, \alpha\zeta, \alpha\zeta^2, \dots, \alpha\zeta^{p-1}$. Thus the splitting field of $f(x)$ over $\mathbb{Q}$ is $\mathbb{Q}(\alpha, \zeta)$. Let G be the Galois group of $f(x)$ over $\mathbb{Q}$. Then each automorphism $\sigma \in G$ is determined by its image of α and ζ as $\mathbb{Q}(\zeta, \alpha)$ is the splitting field of $f(x)$. Since α can be mapped to the roots of its minimal polynomial (which is $x^p - a$), $\sigma(\alpha)$ has at most p choices. Similarly ζ has $p - 1$ choices as the minimal polynomial of ζ over $\mathbb{Q}$ is $x^{p-1} + \cdots + x + 1$. Thus there are possible $p(p-1)$ automorphisms of $\mathbb{Q}(\zeta, \alpha)$ fixing $\mathbb{Q}$. Now $\mathbb{Q}(\zeta, \alpha)$ being the splitting field of the separable polynomial $x^p - a$ the extension is Galois. Therefore

$$|G| = [\mathbb{Q}(\zeta, \alpha) : \mathbb{Q}].$$

Since

$$[\mathbb{Q}(\zeta, \alpha) : \mathbb{Q}] = [\mathbb{Q}(\zeta, \alpha) : \mathbb{Q}(\alpha)][\mathbb{Q}(\alpha) : \mathbb{Q}] = [\mathbb{Q}(\zeta, \alpha) : \mathbb{Q}(\zeta)][\mathbb{Q}(\zeta) : \mathbb{Q}]$$

and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = p$ and $[\mathbb{Q}(\zeta) : \mathbb{Q}] = p - 1$, we have $p, p - 1 \mid |G|$. But p and $p - 1$ are relatively prime implies $p(p - 1) \mid |G|$. Therefore, $|G| \geq p(p - 1)$. But since we saw that $|G| \leq p(p - 1)$, it follows that $|G| = p(p - 1)$. Thus all the maps defined above belong to G . Now let H be the group of transformations of $\mathbb{Z}/(p)$ of the form $y \rightarrow ky + l$. Then define $\Phi : G \rightarrow H$ by $\Phi(\sigma) = \eta$ where $\eta(y) = ky + l$ where $\sigma(\zeta) = \zeta^k$ and $\sigma(\alpha) = \alpha\zeta^l$. Then it is easy to see that Φ is an isomorphism.

Problem 22: Let E/F be the cyclotomic field of the p th roots of unity over the field F of characteristic 0. Show that E can be imbedded in a field K which has a root tower over F such that the integers n_i are primes $[F_{i+1} : F_i] = n_i$. Call such a root tower *normalized*.

Solution: We induct on the degree of the polynomial $x^p - 1$. If $p = 2$, then $E = F$. Now assume that for every $k < p$, a splitting field of $x^k - 1$ can be imbedded into a field K which has a normalized root tower over F and contains all the n th roots of unity for $n \leq k$. Let $K' = K(\zeta)$ where ζ is a primitive p th root of unity. Since the Galois group of the cyclotomic field of order p over a field F of characteristic 0 is abelian, $G = \text{Gal}(K(\zeta)/K)$ is abelian. Therefore G is solvable with a composition series

$$G = G_1 \triangleright G_2 \triangleright \cdots \triangleright G_{k+1} = 1$$

such that G_i/G_{i+1} is cyclic of prime order p_i . Corresponding to G_i , let $\text{Inv } G_i = H_i$. Then by Galois correspondence $[H_{i+1} : H_i] = p_i$. Also since $[K(\zeta) : K] < p$, $p_i < n \forall i$. Therefore, $H_i \supset K$ contains all the p_i th roots of unity by induction hypothesis. Thus $H_{i+1} = H_i(d_i)$ with $d_i^{p_i} \in H_i$ for $i = 1, 2, \dots, k$. Thus $K(\zeta)/K$ has a normalized root tower and by induction hypothesis, K/F has a normalized root tower. Hence $K(\zeta)/F$ has a normalized root tower and $K(\zeta)$ contains the cyclotomic field of the p th roots of unity over F .

Problem 23: Obtain normalized root towers of $\mathbb{Q}$ of the cyclotomic fields of 5th and of 7th roots of unity.

Solution: Imitate the argument in the solution of the above problem.

Problem 24: Prove that, if $f(x) = 0$ has a solvable Galois group over a field F of characteristic 0, then its splitting field can be imbedded in an extension field which has a normalized root tower over F .

Solution: Let E be the splitting field of $f(x)$ over F . Now by the above exercise, F can be imbedded in a field L which contains all the i th roots of unity for $1 \leq i \leq n$ and has a normalized root tower where $n > [E : F]$. Let K be the splitting field of $f(x)$ over L .

Then $E \subseteq K$. By the lemma 4 (section 4.7) of Jacobson, $\text{Gal}(K/L) \leq \text{Gal}(E/F)$. Since $\text{Gal}(E/F)$ is solvable, $\text{Gal}(K/L)$ is solvable (subgroup of a solvable group is solvable.) Then $G = \text{Gal}(K/L)$ has a composition series with prime order composition factors. Correspondingly we get a root tower K/L say

$$H_1 = L \subseteq H_2 \subseteq \cdots \subseteq H_k = K$$

and $[H_{i+1} : H_i] = p_i$, prime. Also $p_i \leq |\text{Gal}(K/L)| \leq |\text{Gal}(E/F)| = [E : F] < n$. Therefore by our assumption, L contains all p_i th roots of unity. By the lemma 3 (section 4.7), $H_{i+1} = H_i(d_i)$ with $d_i^{p_i} \in H_i$. Therefore we get a normalized root tower K/L . Also we have a normalized root tower K/L . Also we have a normalized root tower L/F . Thus we have a normalized root tower K/F in which E is imbedded.

Problem 25: Let E be a splitting field over F of $f(x) \in F[x]$. Show that E is normal over F .

Solution: Let $g(x) \in F[x]$ be irreducible with a root s in E . We want to show that E is normal over F , i.e., we want to show that E contains a splitting field of $g(x)$. Let K be a splitting field of $g(x)$ over E . i.e., $K = E(s_1, \dots, s_m)$ where $g(x) = \prod_1^m (x - s_i) \in K[x]$. Since s_i ($2 \leq i \leq m$) has the same minimum polynomial $g(x)$ over F , we have an isomorphism of $F(s_1)$ to $F(s_i)$ over F sending s_1 to s_i . This can be extended to an isomorphism η of $E(s_1)$ onto $E(s_i)$. Since both of these are splitting field over $F(s_1)$ and $F(s_i)$ respectively of $f(x)$. Then $\eta(E(s_1)) = E(s_i)$ and since $s_1 \in E$, $E(s_1) = E$. Hence $[E : F] = [E(s_i) : F] = [E(s_i) : E][E : F] \Rightarrow [E(s_i) : E] = 1 \Rightarrow s_i \in E \forall i = 1, 2, \dots, m$. Thus all the roots of $g(x)$ lie in E . Hence E is normal over F .

6 The Galois Group as a Permutation Group of the Roots

In the following 6 problems we assume that the characteristic of the base field is $\neq 2$, $f(x) = x^4 - a_1x^3 + a_2x^2 - a_3x + a_4$ has distinct roots r_1, r_2, r_3, r_4 , $E = F(r_1, r_2, r_3, r_4)$, $G = \text{Gal}(E/F)$, and G_f is the corresponding permutation group of the roots.

Problem 26: Show that the subfield of E/F of invariants under $G_f \cap V$ is $F(t_1, t_2, t_3)$ where $t_1 = r_1r_2 + r_3r_4$, $t_2 = r_1r_3 + r_2r_4$, and $t_3 = r_1r_4 + r_2r_3$.

Solution: We want to show that $\text{Inv}(G_f \cap V) = F(t_1, t_2, t_3/F)$. If $\gamma \in G_f \cap V$, then $\gamma(t_i) = t_i \forall i$. Now if $\gamma \in \text{Gal}(F(t_1, t_2, t_3)/F)$, then $\gamma(t_i) = t_i \forall i$. Therefore we can infer that $\gamma(1), \gamma(2) \in \{a, b\}$ and $\gamma(3), \gamma(4) \in \{c, d\}$ where $a, b, c, d \in \{1, 2, 3, 4\}$ and are distinct. Therefore $\gamma \in G_f \cap V$.

Problem 27: Let $g(x) = (x - t_1)(x - t_2)(x - t_3)$ (this polynomial is called the *resolvent cubic* of the quartic $f(x)$). Verify that

$$g(x) = x^3 - b_1x^2 + b_2x - b_3$$

where

$$b_1 = a_2, \quad b_2 = a_1a_3 - 4a_4, \quad b_3 = a_1^2a_4 + a_3^2 - 4a_2a_4$$

and that $f(x)$ and $g(x)$ have the same discriminant.

Solution: We have

$$f(x) = (x - t_1)(x - t_2)(x - t_3) = (x - (r_1r_2 + r_3r_4))(x - (r_1r_3 + r_2r_4))(x - (r_1r_4 + r_2r_3)).$$

If $g(x) = x^3 - b_1x^2 + b_2x - b_3$, then we know that $b_1 = t_1 + t_2 + t_3 = r_1r_2 + r_3r_4 + r_1r_3 + r_2r_4 + r_1r_4 + r_2r_3 = a_2$. Comparing the coefficients, we also get other equalities. Now if d_f denote the discriminant of f and d_g the discriminant of g . Then

$$d_f = (r_1 - r_2)^2(r_1 - r_3)^2(r_1 - r_4)^2(r_2 - r_3)^2(r_2 - r_4)^2(r_3 - r_4)^2$$

and

$$d_g = (t_1 - t_2)^2(t_1 - t_3)^2(t_2 - t_3)^2 = (r_1 - r_4)^2(r_2 - r_3)^2(r_1 - r_3)^2(r_2 - r_4)^2(r_1 - r_2)^2(r_3 - r_4)^2 = d_f.$$

Problem 28: Show that the transitive subgroups of S_4 are (i) S_4 , (ii) A_4 , (iii) $V = \{1, (12)(34), (13)(24), (14)(23)\}$, (iv) $C = \{1, (1234), (13)(24), (1432)\}$ and its conjugates, (v) $D = V \cup \{(12), (34), (1423), (1324)\}$ which is a Sylow 2-group and its conjugates.

Solution: All the subgroups of S_4 given in the question are obviously transitive (Note that conjugate of a transitive subgroup is transitive). For, If G is a transitive subgroup of S_4 and $\sigma \in S_4$, $x, y \in \{1, 2, \dots, n\}$, then since G is a transitive subgroup of S_4 , there exists $g \in G$ such that $g(\sigma^{-1}(x)) = \sigma^{-1}(y)$. Then $(\sigma g \sigma^{-1})(x) = y$. Therefore $\sigma G \sigma^{-1}$ is a transitive subgroup of S_4 .

Now we find all the transitive subgroups of S_4 . Since G is transitive, $|O_x| = 4$, and by orbit stabilizer theorem, $|O_x| \mid |G|$ where O_x is the orbit of x under the action of G . Then $4 \mid |G| \Rightarrow |G| = 4, 8, 12, \text{ or } 24$. If $|G| = 24$, then $G = S_4$. If $|G| = 12$, then $G = A_4$. For, if $|G| = 8$, then $S_4/G \cong \mathbb{Z}_2$, abelian $\Rightarrow G \supset [S_4, S_4] = A_4 \Rightarrow G = A_4$. If $|G| = 4$, then D given in the question is a transitive subgroup. Also, since D is a Sylow 2-subgroup of S_4 , all other 8 order subgroups will be conjugates of D .

Now if $|G| = 4$ and G is cyclic, then we claim that G is a conjugate of C . This is because G is cyclic, $G = \langle (abcd) \rangle$, $a, b, c, d \in \{1, 2, 3, 4\}$ and are distinct. Then $(abcd)$ and (1234) have the same cycle type implies they are conjugates in S_4 . This $gCg^{-1} = G$ for some $g \in S_4$.

Now if $|G| = 4$ and G is isomorphic to Klein 4-group, then we claim that $G = B$. Every non-identity element of G will be of the form (ab) or $(ab)(cd)$. If $(ab) \in G$, then

since G is transitive, (c, d) or $(ab)(cd) \in G$. In either case, $G = \{1, (ab), (cd), (ab0)(cd)\}$ which is not a transitive group as a cannot be mapped to c . Therefore we must have $G = \{1, (ab)(cd), (ac)(bd), (ad)(bc)\} = V$ (this is because elements of V are the only elements of the form of elements of G).

Problem 29: Show that the Galois group G_g of $g(x) = 0$ is isomorphic to $G_f/(G_f \cap V)$. Assume $f(x)$ irreducible and verify that, if (i) $G_f = S_4$ then G_g is of order 6, (ii) $G_f = A_4$, G_g is of order 3, (iii) $G_f = V$, $G_g = 1$, (iv) $G_f = C$ or one of its conjugates, then G_g is of order 2, (v) $G_f = D$ or one of its conjugates, then G_g is of order 2.

Solution: We know that if $E = F(r_1, r_2, r_3, r_4)$, then the subfield of E/F of invariants under $G_f \cap V$ is $F(t_1, t_2, t_3)$. Then by Galois correspondence, $G_f/(G_f \cap V) = \text{Gal}(F(t_1, t_2, t_3)/F)$. But $F(t_1, t_2, t_3)$ is the splitting field of $g(x)$ over F . Thus again by Galois correspondence, $G_g \cong G_f/(G_f \cap V)$. The remaining part of the exercise follows trivially from a cardinality argument given by the equation $|G_g| = |G_f|/|(G_f \cap V)|$.

Problem 30: Prove that if G_g is of order 2, then $G_f \cong D$ or $G_f \cong C$ according as $f(x)$ is or is not irreducible in $F(\sqrt{d})$, d the discriminant of $f(x)$.

Solution: In this question, we assume that $f(x)$ is irreducible over F . We have $G_g \cong G_f/(G_f \cap V)$. If $|G_g| = 2$, then $|G_f| = 2|G_f \cap V|$. Then by a cardinality argument, we see that G_f cannot be S_4, A_4 and V or their conjugates. Since $f(x)$ is irreducible, G_f is transitive. Therefore $G_f \cong C$ or $G_f \cong D$.

Assume $f(x)$ is irreducible in $F(\sqrt{d})$. If $G_f \cong C$, then $\text{Gal}(E/F(\sqrt{d}))$ is also transitive since f is irreducible in $F(\sqrt{d})$. Then $\text{Gal}(E/F(\sqrt{d}))$ is C since there is no proper subgroup of C which is transitive. This implies $\text{Gal}(F(\sqrt{d})) \cong G_f \Rightarrow F(\sqrt{d}) = F \Rightarrow \sqrt{d} \in F \Rightarrow d$ is a square of an element in F . Then by theorem 4.13 (section 4.8), $C \cong G_f \leq A_4$. But A_4 does not have a cyclic subgroup of order 4. This is a contradiction. Therefore, $G_f \cong D$.

Now assume that $f(x)$ is not irreducible over $F(\sqrt{d})$. Suppose $G_f \cong D$, then by theorem 4.13, $\text{Gal}(E/F(\sqrt{d})) = G_f \cap A_4 = V$. Since V is transitive, this implies $f(x)$ is irreducible over $F(\sqrt{d})$ which is a contradiction. Therefore we have $G_f \cong C$.

Problem 31: Determine the Galois group of $x^4 + 3x^3 - 3x - 2 = 0$ over $\mathbb{Q}$.

Solution: The resolvent cubic of $f(x)$ is given by $g(x) = x^3 - x + 9$. Reducing $g(x)$ modulo 2, we get $\bar{g}(x) = x^3 + x + 1$ which has no roots in $\mathbb{Z}_2 \Rightarrow \bar{g}(x)$ is irreducible over $\mathbb{Z}_2 \Rightarrow g(x)$ is irreducible over $\mathbb{Q}$ (mod p test). Therefore G_g is transitive subgroup of S_3 . Now the discriminant of g is given by $d = -2197$. This is obviously not a square in $\mathbb{Q}$. Then $G_g = S_3$ by the remarks succeeding theorem 4.14. Now we have $G_g \cong G_f/(G_f \cap V)$. Since $G_f \cap V \leq V$, $|G_f \cap V| = 1, 2$ or 4 . If $|G_f \cap V| = 2$, then since $|G_g| = |S_3| = 6$, $|G_f| = 2 \times 6 = 12 \Rightarrow G_f = A_4$. But then $G_f \cap V = V \Rightarrow |G_f \cap V| = 4$, contradiction.

If $|G_f \cap V| = 1$, then $G_g \cong G_f \cong S_3$. Since G_f is not a transitive subgroup of S_4 , by theorem 4.14, $f(x) \in F[x]$ is reducible. Note that $f(x)$ does not have any rational roots by rational root theorem. Since $f(x)$ is reducible, there exists $a, b, c, d \in \mathbb{Q}$ such that

$$f(x) = (x^2 + ax + b)(x^2 + cx + d).$$

Let $h(x) = x^2 + ax + b$, $l(x) = x^2 + cx + d$ and r_1, r_2 be the roots of $h(x)$ and r_3, r_4 the roots of $l(x)$. Then if $\sigma \in G_f$, $\sigma(r_1), \sigma(r_2) \in \{r_1, r_2\}$ and $\sigma(r_3), \sigma(r_4) \in \{r_3, r_4\} \Rightarrow \sigma$ is not a 3-cycle in cycle representation for any $\sigma \in G_f$. But this is a contradiction to the fact that $G_f = S_3$ as it contains 3-cycles. Therefore, $G_f \cap V = 1$ is also not possible. Thus $|G_f \cap V| = 4 \Rightarrow |G_f| = 4 \times 6 = 24 \Rightarrow G_f \cong S_4$.

Problem 32: Let H be a normal subgroup $\neq 1$ of a transitive subgroup G of S_n of transformations of $\{1, 2, \dots, n\}$. Show that all H -orbits have the same cardinality. Hence show that if $n = p$ is a prime, then H is transitive.

Solution: Let $O_x^H = \{h.x \mid h \in H\}$. Let H_x denote the stabilizer of $x \in H$, i.e., $H_x = \{h \in H, hx = x\}$. Note that $H_x = G_x \cap H$. Let $x, y \in \{1, 2, \dots, n\}$. Since G is transitive, there exists $g \in G$ such that $g.x = y$. Now $H_y = G_y \cap H = gG_xg^{-1} \cap H = gG_xg^{-1} \cap gHg^{-1} = g(G_x \cap H)g^{-1} = gH_xg^{-1}$ implies $|H_y| = |H_x|$. Then by orbit stabilizer theorem, since $|H|/|O_x^H| = |H_x|$, we get $|O_x^H| = |O_y^H|$.

Let $n = p$. Since all the H -orbits have the same cardinality, $p = k|O_{x_i}^H|$ for distinct orbits $O_{x_i}^H$, $i = 1, \dots, m$ for some $k \geq 1$. Since H contains a non-identity element, $|O_{x_i}^H| > 1$ for some x_i , in fact this is true for all i . This implies that $|O_{x_i}^H| = p$ (p is a prime). $\Rightarrow i = 1. \Rightarrow$ therefore only 1 H -orbit, i.e., H is transitive.

Problem 33: Let p be a prime and let L be the group of all transformations of $\mathbb{Z}_p$ of the form $x \rightarrow ax + b$, $a \neq 0$ including all the translations $x \rightarrow x + b$. Show that the translations $x \rightarrow x + b$, $b \neq 0$, are the only transformations in L without fixed points and hence that these are the only transformations in L whose cycle representations are p -cycles.

Solution: Define a homomorphism $\phi : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ by $\phi(x) = (a - 1)x$ where a fixed number among $\{1, 2, \dots, p - 1\}$. Then ϕ is an isomorphism if $a \neq 1$. Therefore given $b \in \{0, \dots, p - 1\}$, $\exists x \in \mathbb{Z}_p$ such that $(a - 1)x = -b \Rightarrow ax + b = x$. i.e., all the translations $x \rightarrow ax + b$ with $a \neq 1$ has a fixed point. Now consider the translations $x \rightarrow x + b$. If $b = 0$, then the translation has a fixed point. If $b \neq 0$, then if $x + b = x$, then $b = 0$ which is absurd. Thus the translations of the form $x \rightarrow x + b$, $b \neq 0$ are the only transformations in L without fixed points.

Now to prove the last part, suppose $x \rightarrow ax + b$, $b \neq 0$ has a p -cycle representation, then this implies that $x \rightarrow ax + b$ does not have a fixed point as p -cycles will not have fixed points. But this is a contradiction. Therefore, $x \rightarrow x + b$, $b \neq 0$ are the only

transformations in L whose cycle representations are p -cycles.

7 Constructible Regular n -gons. Cyclotomic fields over $\mathbb{Q}$

Problem 34: Use the Möbius inversion formula to prove that

$$\lambda_n(x) = \prod_{d|n} (x^d - 1)^{\mu(n/d)}.$$

Solution: Let $\mu(n)$ denote the Möbius function of positive integers. Then Möbius inversion formula is given by

$$f(x) = \sum_{d|n} \mu(n/d) g(d)$$

where $f(n)$ is a function of positive integers with values in an abelian group and $g(n) = \sum_{d|n} f(d)$. We know that $x^n - 1 = \prod_{d|n} \lambda_d(x)$. Here, take $f(n) = \lambda_n(x) \in \mathbb{R}[x]^*$ (group of rational functions under point wise multiplication). Then

$$g(n) = \sum_{d|n} f(d) = \prod_{d|n} \lambda_n(x) = x^n - 1.$$

Therefore by Möbius inversion formula ,

$$\lambda_n(x) = f(n) = \sum_{d|n} \mu(n/d) g(d) = \prod_{d|n} (x^d - 1)^{\mu(n/d)}.$$

Problem 35: Let $f(x)$ have distinct roots $r_1, r_2, \dots, r_n$. Show that the discriminant

$$d = \prod_{1 \leq i < j \leq n} (r_i - r_j)^2 = (-1)^{n(n-1)/2} \prod_{i=1}^n f'(r_i)$$

f' , the derivative of f . Let $\lambda_p(x) = x^{p-1} + x^{p-2} + \dots + 1$, p a prime. Differentiate $x^p - 1 = (x - 1)\lambda_p(x)$ to obtain

$$px^{p-1} = \lambda_p(x) + (x - 1)\lambda_p'(x).$$

Show that the discriminant of the cyclotomic polynomial $\lambda_p(x)$ is

$$d = (-1)^{p(p-1)/2} p^{p-2}.$$

Solution: Let $f(x) = (x - r_1) \cdots (x - r_n)$. When $n = 2$, the above equation is obviously true. Assume the above equation for discriminant in terms of derivative is true for $g(x) = (x - r_1) \cdots (x - r_{n-1})$ where $r_1, \dots, r_{n-1}$ are distinct roots of g . Then the discriminant of g . $d_g = (-1)^{(n-1)(n-2)/2} \prod_{i=1}^{n-1} g'(r_i)$. Now let $f(x) = (x - r_n)g(x)$ where

r_n is distinct from $r_1, \dots, r_{n-1}$. Then $f'(x) = g'(x)(x - r_n) + g(x)$.

$$d_f = d_g(r_1 - r_n)^2 \cdots (r_{n-1} - r_n)^2. \quad (4)$$

Now,

$$\begin{aligned} \prod_1^n f'(r_i) &= \prod_1^{n-1} g'(r_i)(r_i - r_n)g(r_n) \\ &= (-1)^{(n-1)(n-2)/2} d_g(r_1 - r_n) \cdots (r_{n-1} - r_n)(r_{n-1} - 1) \cdots (r_n - r_{n-1}) \\ &= (-1)^{(n-1)(n-2)/2} (-1)^{n-1} d_g(r_1 - r_n)^2 \cdots (r_{n-1} - r_n)^2 \\ &= (-1)^{n(n-1)/2} d_g(r_1 - r_n)^2 \cdots (r_{n-1} - r_n)^2 \end{aligned}$$

$$\Rightarrow d_f = (-1)^{n(n-1)/2} \prod_1^n f'(r_i) \text{ (from equation 4).}$$

Now consider $x^p - 1 = (x - 1)\lambda_p(x)$. Differentiating with respect to x , we get $px^{p-1} = \lambda_p(x) + (x - 1)\lambda'_p(x)$. Let $r_1, \dots, r_{p-1}$ be all the roots of $\lambda_p(x)$. Then from the above equation, $\lambda'_p(r_i) = pr_i^{p-1}/(r_i - 1)$. By the first part of the problem,

$$\begin{aligned} d_{\lambda_p} &= (-1)^{(p-1)(p-2)/2} \prod_1^{p-1} \lambda'_p(r_i) \\ &= (-1)^{(p-1)(p-2)/2} \prod_1^{p-1} \frac{pr_i^{p-1}}{r_i - 1} \\ &= (-1)^{(p-1)(p-2)/2} \frac{p^{p-1}(r_1 \cdots r_{p-1})}{(r_1 - 1) \cdots (r_{p-1} - 1)}. \end{aligned}$$

Since $r_1, \dots, r_{p-1}$ are primitive p th roots of unity, $r_1 \cdots r_{p-1} = 1$. Also we have $\lambda_p(x) = (x - r_1) \cdots (x - r_{p-1}) = x^{p-1} + \cdots + 1 \Rightarrow \lambda_p(1) = (1 - r_1) \cdots (1 - r_{p-1}) = p$. Thus

$$d_{\lambda_p} = (-1)^{(p-1)(p-2)/2} \frac{p^{p-1}}{(-1)^{p-1}p} = (-1)^{(p-1)p/2} p^{p-2}.$$

Problem 36: Let Λ_p be the field of the p th roots of unity over $\mathbb{Q}$ where p is an odd prime. Show that Λ_p has a unique quadratic subfield $E/\mathbb{Q}$ and E is real (subfield of $\mathbb{R}$) or not real according as p has the form $4n + 1$ or $4n + 3$.

Solution: We know that $\Lambda_p = \mathbb{Q}(\zeta)$ where ζ is a primitive p th root of unity and $\text{Gal}(\Lambda_p/\mathbb{Q}) \cong \mathbb{Z}/(p-1)\mathbb{Z}$. Since p is an odd prime, $p-1$ is even and $\mathbb{Z}_{p-1}$ has a unique subgroup H of index 2. Let $\text{Inv } H = E$. Then E has index 2 over $\mathbb{Q}$. Also E is the unique quadratic extension by Galois correspondence. We claim that $E = \mathbb{Q}(\sqrt{d})$ where d is the discriminant of $\lambda_p(x)$. From the previous problem, it is clear that $d = (-1)^{p(p-1)/2} p^{p-2}$ so that $d \in \mathbb{Q}$. Therefore $[\mathbb{Q}(\sqrt{d}) : \mathbb{Q}] = 1$ or 2 . If it was 1, then $\sqrt{d} \in \mathbb{Q} \Rightarrow \mathbb{Z}_{p-1} \leq A_{p-1}$ (corollary to theorem 4.13 Jacobson). This implies A_{p-1} contains a $(p-1)$ cycle. But

since p is an odd prime, $p - 1$ is even and therefore $p - 1$ cycle is an odd permutation. Therefore $[\mathbb{Q}(\sqrt{d}) : \mathbb{Q}] = 2$. But there is only one quadratic subfield of Λ_p over $\mathbb{Q}$. Thus $E = \mathbb{Q}(\sqrt{d})$. Note that any odd prime is of the form $4n + 1$ or $4n + 3$, $n \geq 0$. $p = 4n + 1$, then $d = (-1)^{(4n+1)(4n+1-1)/2} p^{p-2} = p^{p-2} > 0$. Then $\sqrt{d} \in \mathbb{R}$ and $\mathbb{Q}(\sqrt{d}) \subset \mathbb{R}$. Similarly, if $p = 4n + 3$, then $d < 0$ so that $\sqrt{d} \notin \mathbb{R}$ and $E \not\subset \mathbb{R}$.